

Lanternhill Karaoke & Comedy Lounge Chain

Campus LAN Segmentation, Trunking & Spanning Tree — Case Study No. 137

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1. Business Scenario & Problem

Lanternhill is a chain of three karaoke-and-comedy lounges — Midtown, Riverside, and Warehouse District — interconnected in a triangle over leased metro fibre so they can share one booking and song-catalog system. Every device currently sits on a single flat network, which causes two problems:

- **Broadcast storms during weekend peaks** — too many devices in one broadcast domain, amplified by the physical triangle loop.
- **No security separation** — guests' personal devices can reach the stage-lighting and sound-control systems.

The task is to redesign the switching layer: segment with VLANs, secure the trunks, guarantee a loop-free topology, add cheap link redundancy, and ensure guests can never reach the stage systems.

2. Core Concepts (the “why” behind the design)

Broadcast frames & broadcast domains

Most frames are unicast — a switch learns which device (MAC) sits on which port and forwards only there. A broadcast frame (destination MAC `ffff.ffff.ffff`) must be flooded out every port, because it is how devices find each other (ARP “who has this IP?”, DHCP, discovery). A broadcast domain is the set of devices that hear each other's broadcasts. On a flat network every device is in one giant broadcast domain, so on a busy night hundreds of devices all broadcast into the same “room,” and every device must process every broadcast — that is the noise. VLANs split the one big room into several smaller broadcast domains, so a guest-phone broadcast never reaches the stage gear, and each domain is quieter.

Why a Layer-2 loop is catastrophic (no TTL)

The triangle means each switch connects to the other two — a physical loop. A broadcast frame entering the loop is flooded onward by each switch, and crucially a Layer-2 frame has no TTL (time-to-live) counter — unlike a Layer-3 packet, which expires. So the frame loops forever and multiplies at every switch, saturating the network within seconds. That is a broadcast storm, and it is exactly the “dies on busy weekends” complaint. The loop is not removed physically (the redundancy is wanted); instead STP blocks one path logically while keeping it as hot standby.

Spanning Tree Protocol (STP)

STP finds loops and blocks just enough ports to break every loop, keeping all switches reachable and the blocked links as standby. Switches elect a root bridge by lowest bridge priority (ties broken by lowest MAC). Default priority is 32768 and must be set in multiples of 4096; the VLAN ID is added as the “sys-id-ext” (so VLAN 61 at priority 4096 shows as 4157). The root's ports are all Designated/Forwarding; every other switch keeps one Root Port (best path to root) forwarding; one end of each link is Designated/Forwarding; and the port that would complete a loop is put into Blocking.

STP path cost by link speed: Gigabit = 4, FastEthernet = 19 — faster links are “cheaper,” so lower cumulative cost wins the path. An EtherChannel of two Gig links costs 3 (cheaper still).

Cheap redundancy & EtherChannel

The case asks for link redundancy “where it is cheap to do so.” The two parallel Warehouse–Midtown fibres are exactly that: bundling them into one EtherChannel is cheap redundancy. Without bundling, STP would block one of the two parallel links (wasting it). Once bundled, STP sees a single logical link (Port-channel 1), so there is no loop to break — both cables forward together, roughly doubling bandwidth and providing redundancy, and the path cost drops from 4 to 3.

3. Hardware & Design Decisions

Role	Device	Reason
Edge/HQ + routing	SW-Warehouse — Catalyst 3560 (multilayer)	STP root + inter-VLAN routing (SVIs) + ACL, all in one device.
Access switches	SW-Midtown, SW-Riverside — Catalyst 2960	VLAN segmentation and end-device connectivity.
Guest access	Access Point (described) / wired PCs via hub	Guests join Wi-Fi via the AP; wired PCs used to demonstrate port security.
Shared server	Booking/Catalog server on VLAN 91	Central booking + song catalog, reached by all venues via routing.

Multilayer switch vs router-on-a-stick: a router-on-a-stick funnels all inter-VLAN traffic through a single trunk (a bottleneck); a multilayer switch routes in hardware with SVIs and no choke point. A Layer-3 switch was chosen — it also doubles as the STP root and ACL enforcement point, simplifying the topology.

Added VLAN (Objective 1): VLAN 91 “Servers” was added to isolate the shared booking/catalog server from user broadcast domains and to control access to it centrally via ACLs. It exists only at Warehouse (where the server lives) and never crosses a trunk — other venues reach it by routing.

4. VLAN Scheme & Port Assignments

VLAN	Name	Lives at	Subnet / Gateway
51	Stage-Sound	Midtown, Riverside	192.168.51.0/24 · .1
61	Staff-POS	All venues	192.168.61.0/24 · .1
71	Admin	Warehouse	192.168.71.0/24 · .1
81	IT-Mgmt	All venues	192.168.81.0/24 · .1
91	Servers (added)	Warehouse	192.168.91.0/24 · .1
955	Guest	Riverside	192.168.100.0/24 · .1
999	Native (unused)	All trunks	— (anti VLAN-hopping)

Note: the Guest VLAN is numbered 955 but uses the 192.168.100.0/24 subnet — 192.168.955.x is invalid (octets max at 255); the VLAN number and IP subnet are independent.

Port assignments

Switch	Access ports	Trunk / uplink ports
SW-Warehouse (L3, root)	Fa0/1–3 → 61 · Fa0/4–6 → 71 · Fa0/7–8 → 81 · Fa0/9–10 → 91	Gig0/1–2 → EtherChannel (Po1) to Midtown · Fa0/24 → Riverside
SW-Midtown	Fa0/1–3 → 51 · Fa0/4–6 → 61 · Fa0/7–8 → 81	Gig0/1–2 → EtherChannel (Po1) to Warehouse · Fa0/24 → Riverside
SW-Riverside	Fa0/1–3 → 51 · Fa0/4–6 → 61 · Fa0/7–8 → 81 · Fa0/9–10 → 955 (port security)	Fa0/23 → Midtown · Fa0/24 → Warehouse

5. Trunking & Native VLAN 999

All three inter-switch links are 802.1Q trunks carrying VLANs 51, 61, 81, 955 (VLAN 91 stays local to Warehouse). Native VLAN is set to the unused VLAN 999 on every trunk, and must match on both ends.

```
! SW-Warehouse (3560 needs dot1q selected first)
interface range gig0/1 - 2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 51,61,71,81,955
```

Security reason for native VLAN 999 (Objective 2):

802.1Q sends native-VLAN traffic untagged across a trunk. An attacker on an access port can exploit this with double-tagging: a frame carrying two VLAN tags has its outer (native) tag stripped by the first switch and is then forwarded with the inner tag into a VLAN the attacker should not reach — VLAN hopping. By setting the native VLAN to an unused, dead-end VLAN (999) with no access ports and no real traffic, any frame slipping through on the native VLAN lands in a VLAN that goes nowhere, closing the double-tagging path while the real VLANs stay tagged end to end.

6. Spanning Tree — Root Bridge, Priorities & Port Roles

SW-Warehouse is forced to be the root bridge (lowest priority); Midtown is the backup root; Riverside keeps the default.

Switch	VLAN priority set	Resulting bridge priority
SW-Warehouse (root)	4096	4096 + VLAN-ID (e.g. 4157 for VLAN 61)
SW-Midtown (backup)	8192	8192 + VLAN-ID
SW-Riverside	32768 (default)	32768 + VLAN-ID

```
spanning-tree vlan 51,61,71,81,955 priority 4096 ! Warehouse
spanning-tree vlan 51,61,71,81,955 priority 8192 ! Midtown
```

Resulting port role / state table (verified, after EtherChannel):

Switch	Port	Role	State	Cost
Warehouse (root)	Po1 (Gig0/1–2)	Designated	Forwarding	3
Warehouse (root)	Fa0/24	Designated	Forwarding	19
Midtown	Po1 (Gig0/1–2)	Root	Forwarding	3
Midtown	Fa0/24	Designated	Forwarding	19
Riverside	Fa0/24	Root	Forwarding	19
Riverside	Fa0/23	Alternate	Blocking	19

The blocked port (Riverside Fa0/23, facing Midtown) is what breaks the triangle loop. Note: a green link light means the cable is up, not that the port is forwarding — STP can hold an up link in Blocking (shown amber).

7. EtherChannel (Warehouse ↔ Midtown)

The two parallel Gig links are bundled with LACP (mode active on both ends). Trunk settings must also be applied to the Port-channel so the members and the bundle match.

```
interface range gig0/1 - 2
  channel-group 1 mode active
interface port-channel 1
  switchport mode trunk
  switchport trunk native vlan 999
  switchport trunk allowed vlan 51,61,71,81,955
```

What happens to STP once EtherChannel is active (Objective 5):

Before bundling, STP saw two separate links, treated them as a loop, and blocked one (Gi0/2 was Alternate/Blocking — a wasted link). After bundling, STP sees a single logical interface (Port-channel 1): with no loop there is nothing to block, so both physical cables forward together. The path cost also dropped from 4 to 3, and STP now runs its calculations on Po1 as one unit. Verified with show etherchannel summary showing Po1(SU) with both ports (P).

8. Inter-VLAN Routing (SVIs) & the Guest ACL

Routing is enabled on SW-Warehouse with one SVI per VLAN acting as its gateway. An SVI only comes up (up/up) once at least one active device exists in its VLAN — VLAN 91's SVI stayed up/down until the server was connected.

```
ip routing
interface vlan 61
  ip address 192.168.61.1 255.255.255.0
! ... one SVI per VLAN ...
```

ACL — Guest must never reach Stage-Sound (Objective 3):

```
ip access-list extended GUEST-RESTRICT
deny ip 192.168.100.0 0.0.0.255 192.168.51.0 0.0.0.255
deny ip 192.168.100.0 0.0.0.255 192.168.61.0 0.0.0.255
deny ip 192.168.100.0 0.0.0.255 192.168.71.0 0.0.0.255
deny ip 192.168.100.0 0.0.0.255 192.168.81.0 0.0.0.255
deny ip 192.168.100.0 0.0.0.255 192.168.91.0 0.0.0.255
permit ip any any
interface vlan 955
  ip access-group GUEST-RESTRICT in
```

Applied inbound on the Guest SVI, it blocks guest traffic as it enters the router. The required rule is Guest X Stage-Sound (51); the others harden the design so guests reach no internal VLAN while still being permitted outward (to the internet, if an ISP were added). Guests can still reach their own gateway (needed) but cannot route through it to internal hosts.

Internet Edge (ISP) & Giving Every VLAN Internet Access

An ISP router is attached to SW-Warehouse on a dedicated routed port (Fa0/23, no switchport). Warehouse points a default route at the ISP, and the ISP points a route for the private ranges back at Warehouse, so every internal VLAN can reach the internet (represented by a loopback 8.8.8.8 on the ISP). The Guest ACL still blocks guests from internal VLANs while permitting them outward, so guests reach the internet but nothing internal.

SW-Warehouse:

```
interface fa0/23
no switchport
ip address 203.0.113.1 255.255.255.252
no shutdown
ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

ISP router:

```
interface gig0/0
ip address 203.0.113.2 255.255.255.252
interface loopback0
ip address 8.8.8.8 255.255.255.255
ip route 192.168.0.0 255.255.0.0 203.0.113.1
```

Result: any internal PC and the guest PC can ping 8.8.8.8 (internet), while the guest PC remains blocked from internal VLANs by the ACL.

Why routing and not NAT — and what production needs: a Catalyst 3560 multilayer switch does not support NAT (only 6500/5500-class switches do; the command is rejected on the 3560). For the lab, static routing with a return route is sufficient to demonstrate guest-internet access and the ACL boundary. In production, private campus addresses must be translated to a public address before reaching the internet, so a dedicated edge router or firewall (e.g. a Cisco ASA) sits at the internet boundary to perform NAT/PAT — the L3 switch handles internal inter-VLAN routing while the edge device handles NAT and the public boundary.

9. Port Security & the Guest-Port Demonstration

Riverside's guest access ports (Fa0/9–10) allow a maximum of 2 learned MAC addresses with violation action restrict and sticky learning.

```
interface range fa0/9 - 10
switchport mode access
switchport access vlan 955
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky
```

Why “restrict” over “shutdown” (Objective 6):

In a public lounge, guest devices connect and disconnect constantly, so violations happen routinely and innocently. With shutdown, the first violation err-disables the port completely, cutting off all guests on it until IT manually re-enables it — an unacceptable support burden. Restrict instead drops only the offending device's traffic and logs the event, while legitimate guests keep working. Same protection against MAC flooding and rogue devices, without self-inflicted outages.

The hub + three-PC demonstration

A hub with three PCs was connected to one guest port (Fa0/9) to force a violation. The port learned the first two MACs (sticky) and restricted the third; the violation counter climbed and PSECURE_VIOLATION messages logged. Verified with show port-security interface fa0/9 (Total MAC Addresses: 2, Security Violation Count rising).

The three PCs returned three different errors — each reveals a different point of failure:

- **“Destination host unreachable”** — the PC whose traffic reached the router and was blocked by the Layer-3 ACL (correct ACL behaviour).
- **“Request timed out”** — the third PC, dropped at the switch port by Layer-2 port security; its frames never reached the router, so nothing replies.

- **“Ping could not find host”** — a local PC issue (a pinged name/typo or missing gateway), not a network block; it failed before any packet left.

This neatly shows two security layers catching traffic at two different points — port security at Layer 2 and the ACL at Layer 3.

Why a hub — and not a switch — for the demonstration

Both a hub and a switch pass the PCs' frames through with their original source MACs, so either would let Fa0/9 see three MACs. A hub is preferred because it is a dumb Layer-1 repeater that generates nothing of its own — so the port sees exactly the three PC MACs, a clean count. A switch would additionally emit its own STP/CDP frames from its own MAC, adding a stray MAC to the count and making the output messier. (A Wi-Fi router doing NAT would do the opposite problem — hide all devices behind one MAC — defeating the count entirely.) For reference: a hub is a single collision and broadcast domain; a switch gives each port its own collision domain — which is why switches replaced hubs.

10. VTP Mode Recommendation (Memo)

Recommendation: configure all three switches in VTP Transparent mode.

This is a small, stable network — three switches, few VLANs that rarely change — so the automatic VLAN propagation of Server/Client mode saves little, while introducing a real risk. The VTP configuration-revision hole means a switch with a higher revision number added to the domain (even an old lab unit) can overwrite the VLAN database on every other switch, instantly wiping the production VLANs. For a business where the stage-sound, POS and admin VLANs must never vanish mid-service, that single-point catastrophe is unacceptable.

Transparent mode removes that risk: each switch keeps its own local VLANs, unaffected by others' advertisements, while still forwarding VTP messages along the trunks. Its configuration revision stays at 0, so it can never be overwritten. The only cost — defining VLANs manually on each switch — is trivial at three switches. Trade-off: we give up automatic sync (minor) for immunity to accidental VLAN-database wipeouts (major). For a small, change-averse, service-critical network, that favours Transparent.

Verified: show vtp status → VTP Operating Mode: Transparent, Configuration Revision: 0.

11. Link-Failure Convergence Report (Objective 8)

Scenario: a simulated fibre cut on the Warehouse↔Riverside link (shut SW-Riverside Fa0/24).

- **Before:** Riverside reaches the root via Fa0/24 (Root Port, Forwarding); Fa0/23 (to Midtown) is Alternate/Blocking.
- **On failure:** Fa0/24 goes down; Riverside loses its path to the root. STP detects the loss and transitions the blocked Fa0/23 through Blocking → Listening → Learning → Forwarding, promoting it to the new Root Port.
- **After:** Riverside reaches the root via Midtown (Fa0/23 = Root/Forwarding). Connectivity restored, still loop-free. New path: Riverside → Midtown → EtherChannel → Warehouse.
- **Convergence time:** ~30 s on classic STP (two forward-delay periods through Listening and Learning). Rapid-PVST would converge in ~1–2 s — a recommended improvement.

```
interface fa0/24
shutdown      ! simulate the cut; wait ~30s; show spanning-tree vlan 61
no shutdown   ! restore
```

12. Troubleshooting Log (issues met & resolved)

1. **Access ports defaulted to VLAN 1.** Creating a VLAN does not place any port in it. End-device ports initially sat in VLAN 1, so PCs could not reach their gateways. Fixed by assigning each access port with `switchport access vlan X`; `show vlan brief` catches this.
2. **EtherChannel would not bundle ("vlan mask is different").** Members showed (I)/(SD) because the Port-channel's trunk settings did not match the physical ports. Fixed by applying the same trunk config (native 999, allowed list) to Port-channel 1, or clearing the members (no channel-group) and re-adding them so they inherit cleanly. EtherChannel only forms when every member matches.
3. **"LACP not enabled on remote port."** Only one end had channel-group configured; LACP needs both ends set to mode active. Resolved once both switches were configured.
4. **3560 trunk needs encapsulation first.** The multilayer switch requires `switchport trunk encapsulation dot1q` before `switchport mode trunk`; the 2960s are dot1q-only and reject that command (expected, not an error).
5. **SVI stayed up/down.** VLAN 91's SVI did not come up until the server was connected and active in VLAN 91 — an SVI needs at least one live port in its VLAN.
6. **Invalid Guest IP.** 192.168.955.x is invalid (octet > 255); the Guest VLAN (955) uses the 192.168.100.0/24 subnet instead.
7. **Ping error messages.** "Could not find host" = typed a name/typo (fails before sending). "Request timed out" = dropped by port security at the port. "Destination host unreachable" = blocked by the ACL at the router. The first ping often drops during ARP, then succeeds.
8. **PT command syntax.** `show run interface X` was rejected; used `show running-config` instead.

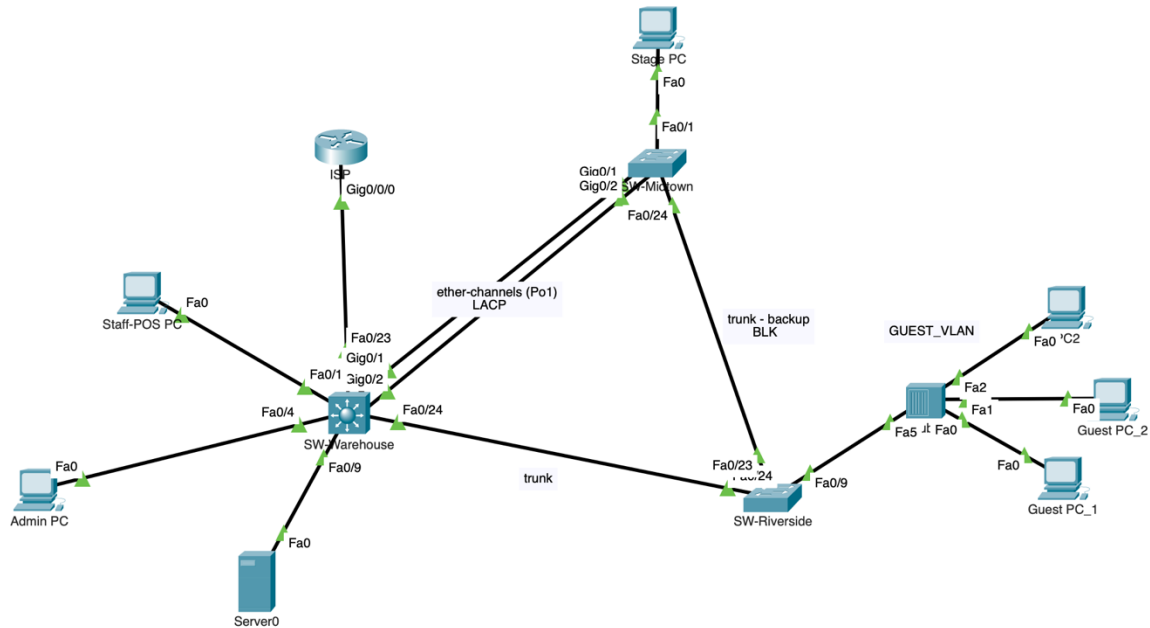
13. Concepts Demonstrated

VLAN segmentation · 802.1Q trunking · native-VLAN hardening (999) · inter-VLAN routing via SVIs · Layer-3 multilayer switching · Spanning Tree (root election, priorities, port roles/states) · STP path cost · broadcast-domain / broadcast-storm theory (no-TTL loop) · LACP EtherChannel (cheap redundancy + cost reduction) · extended ACL policy enforcement · internet edge via static routing (and the real-world NAT boundary) · port security (max-MAC, restrict, sticky) · hub vs switch vs NAT behaviour · VTP mode trade-off · STP convergence & failover.

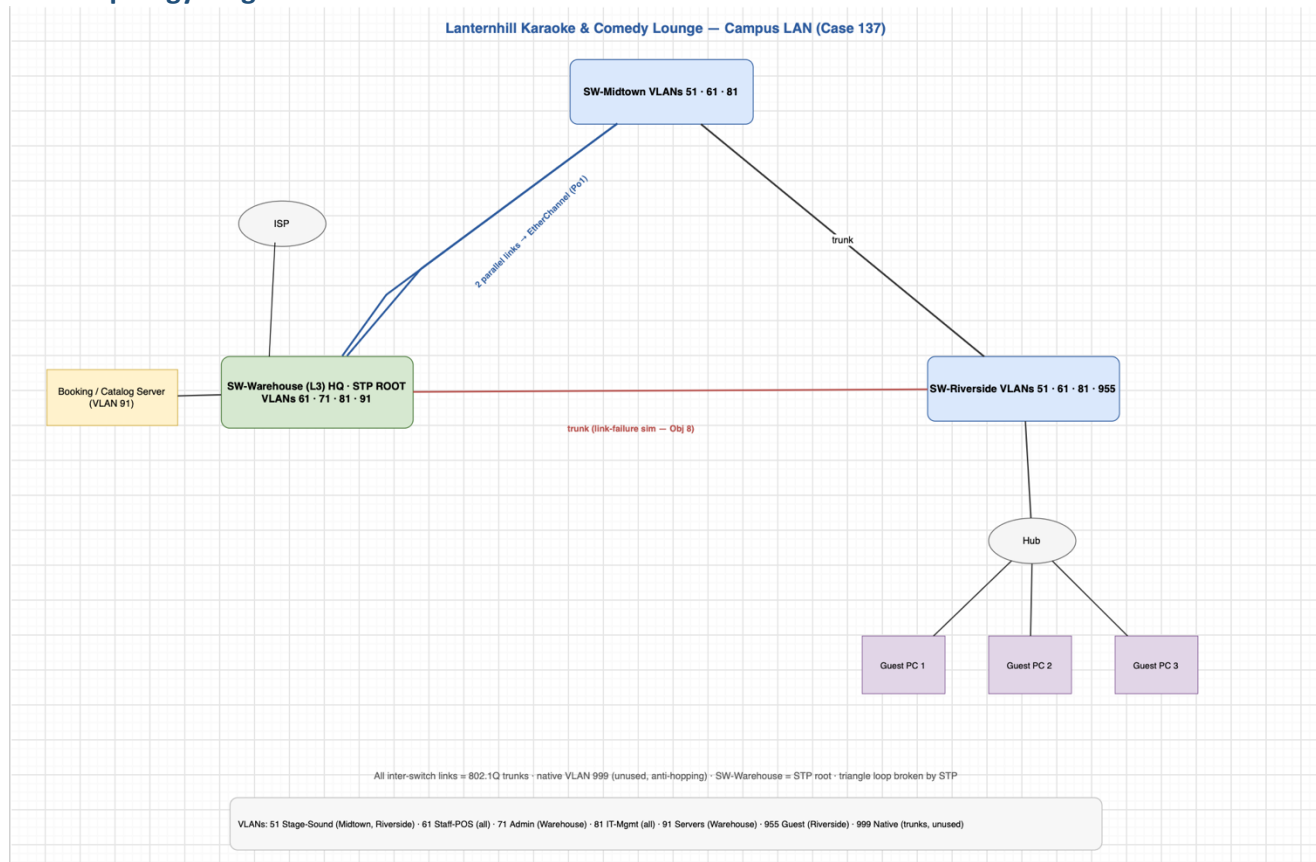
14. Evidence & Screenshots

Insert the captured screenshots in the spaces below.

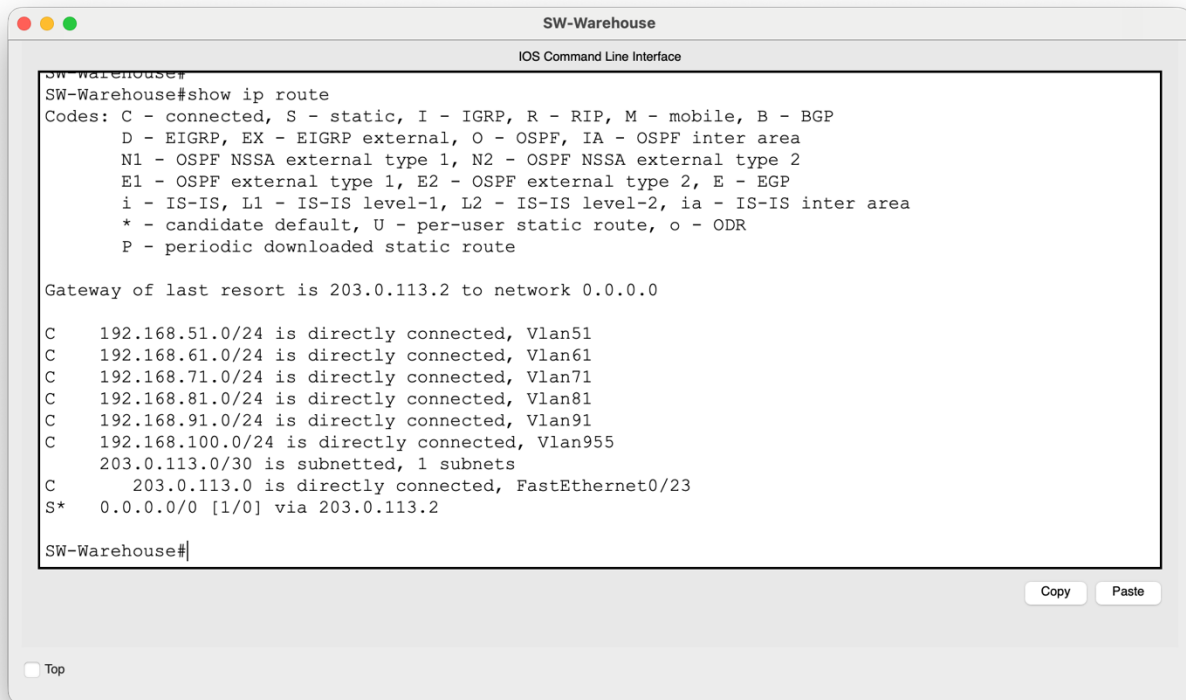
14.1 Final working topology



14.2 Topology diagram



14.3 Routing table (SW-Warehouse)



```
SW-Warehouse#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

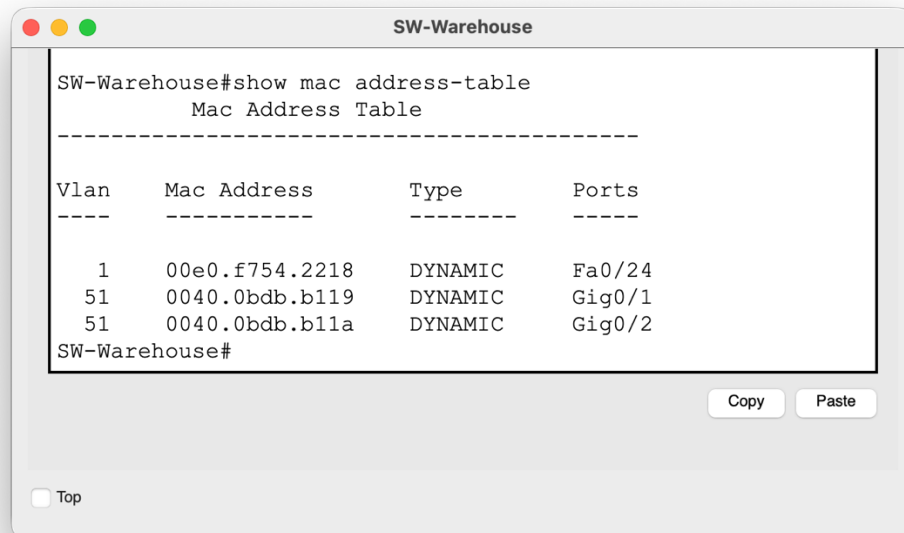
C    192.168.51.0/24 is directly connected, Vlan51
C    192.168.61.0/24 is directly connected, Vlan61
C    192.168.71.0/24 is directly connected, Vlan71
C    192.168.81.0/24 is directly connected, Vlan81
C    192.168.91.0/24 is directly connected, Vlan91
C    192.168.100.0/24 is directly connected, Vlan955
C    203.0.113.0/30 is subnetted, 1 subnets
C      203.0.113.0 is directly connected, FastEthernet0/23
S*   0.0.0.0/0 [1/0] via 203.0.113.2

SW-Warehouse#
```

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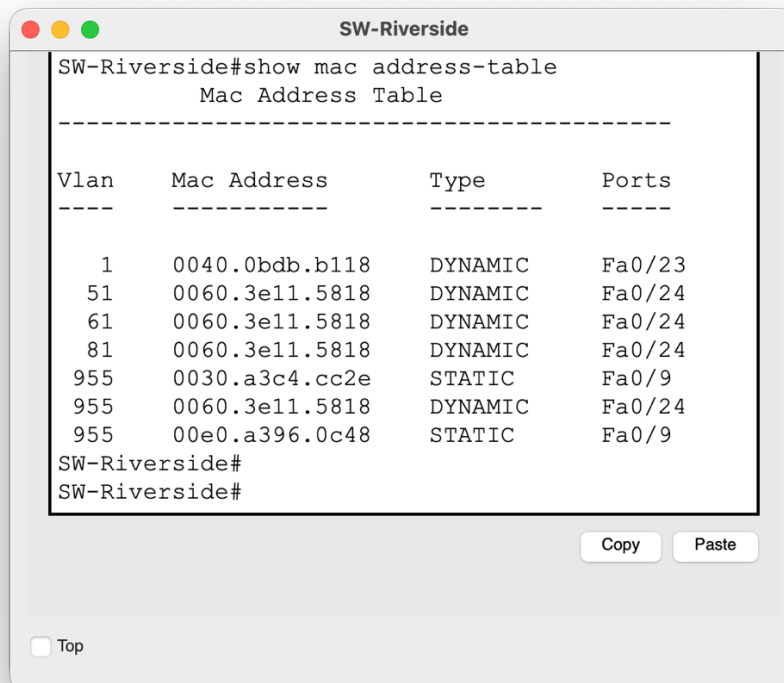
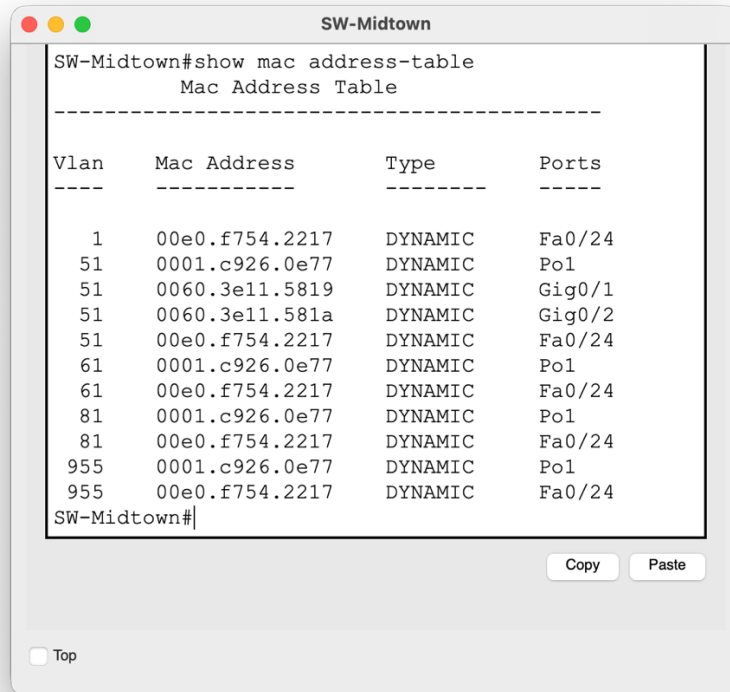
14.4 MAC address tables



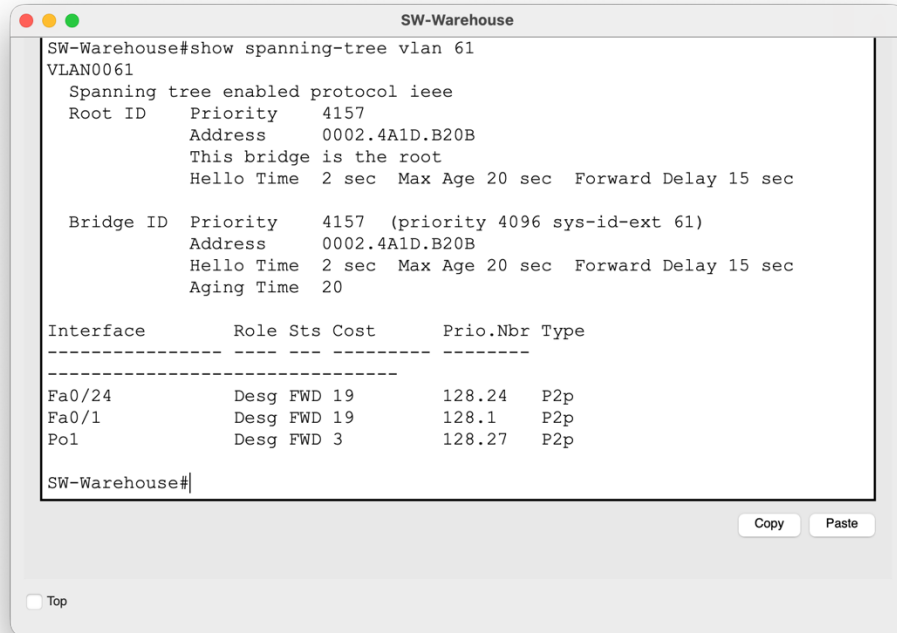
```
SW-Warehouse#show mac address-table
          Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
1       00e0.f754.2218    DYNAMIC   Fa0/24
51      0040.0bdb.b119    DYNAMIC   Gig0/1
51      0040.0bdb.b11a    DYNAMIC   Gig0/2
SW-Warehouse#
```

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14.5 Spanning-tree status & port roles



```
SW-Warehouse#show spanning-tree vlan 61
VLAN0061
  Spanning tree enabled protocol ieee
  Root ID    Priority    4157
            Address     0002.4A1D.B20B
            This bridge is the root
            Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

  Bridge ID  Priority    4157 (priority 4096 sys-id-ext 61)
            Address     0002.4A1D.B20B
            Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
            Aging Time 20

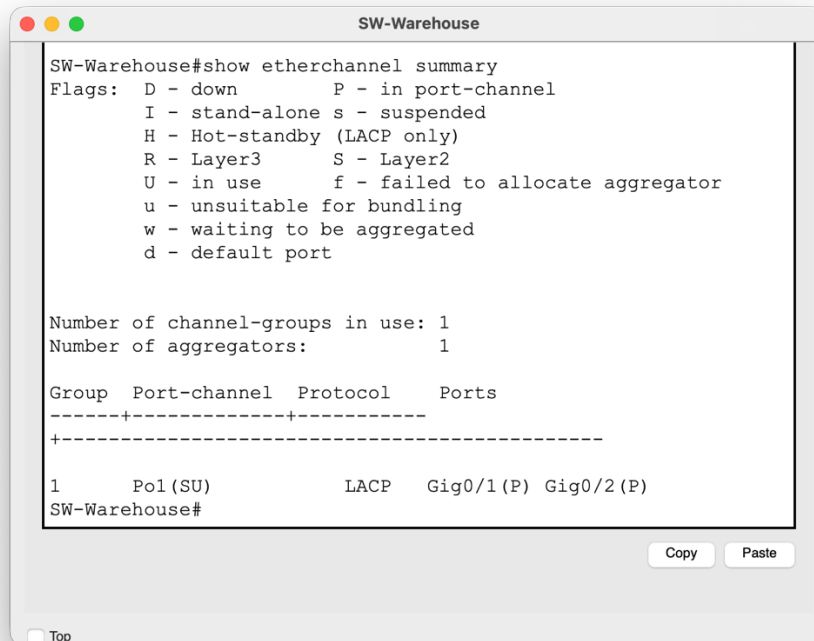
Interface                Role Sts Cost      Prio.Nbr Type
-----
Fa0/24                   Desg FWD 19       128.24   P2p
Fa0/1                    Desg FWD 19       128.1    P2p
Pol                      Desg FWD 3       128.27   P2p

SW-Warehouse#
```

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14.6 EtherChannel summary



```
SW-Warehouse#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

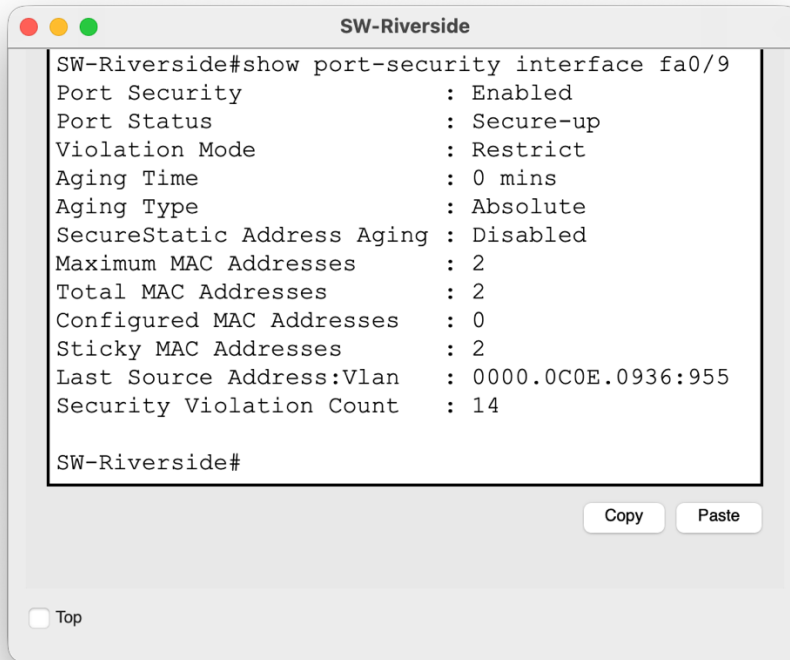
Group  Port-channel  Protocol    Ports
-----
+-----+-----+-----+
1      Pol (SU)      LACP       Gig0/1 (P) Gig0/2 (P)

SW-Warehouse#
```

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14.7 Port security



```
SW-Riverside#show port-security interface fa0/9
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Restrict
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 2
Total MAC Addresses     : 2
Configured MAC Addresses : 0
Sticky MAC Addresses    : 2
Last Source Address:Vlan : 0000.0C0E.0936:955
Security Violation Count : 14

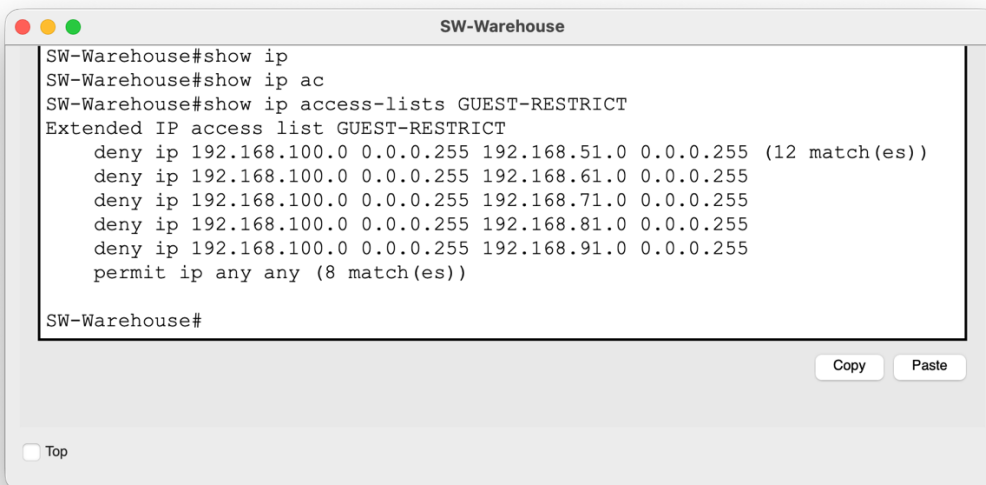
SW-Riverside#
```

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14.8 ACL verification

Command: show ip access-lists GUEST-RESTRICT + Guest→Stage ping failing

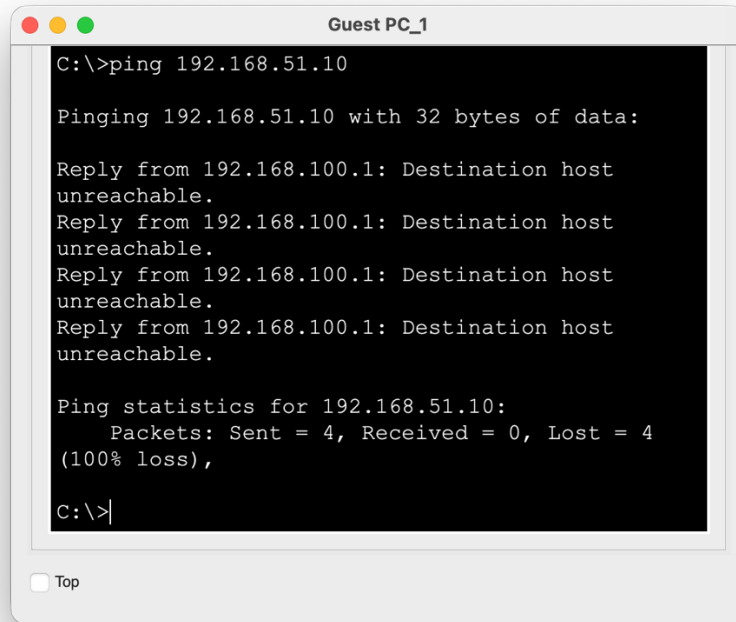


```
SW-Warehouse#show ip
SW-Warehouse#show ip ac
SW-Warehouse#show ip access-lists GUEST-RESTRICT
Extended IP access list GUEST-RESTRICT
deny ip 192.168.100.0 0.0.0.255 192.168.51.0 0.0.0.255 (12 match(es))
deny ip 192.168.100.0 0.0.0.255 192.168.61.0 0.0.0.255
deny ip 192.168.100.0 0.0.0.255 192.168.71.0 0.0.0.255
deny ip 192.168.100.0 0.0.0.255 192.168.81.0 0.0.0.255
deny ip 192.168.100.0 0.0.0.255 192.168.91.0 0.0.0.255
permit ip any any (8 match(es))

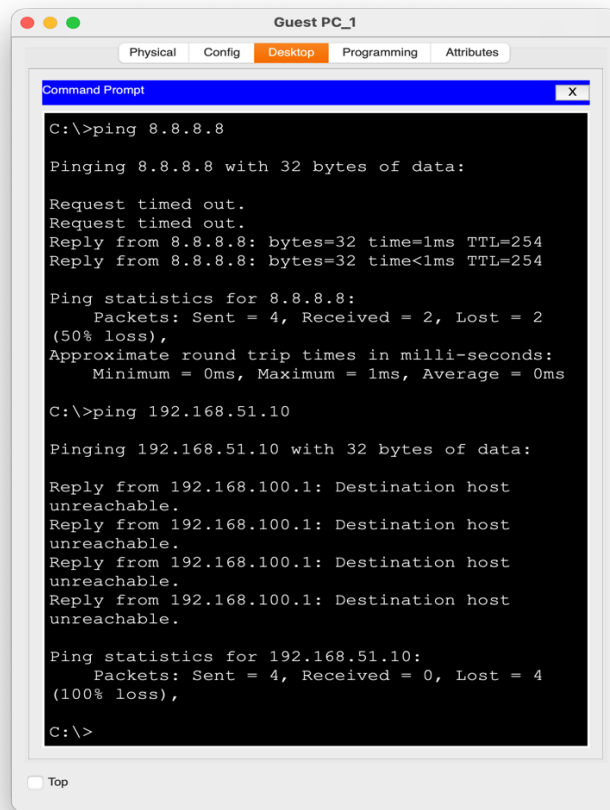
SW-Warehouse#
```

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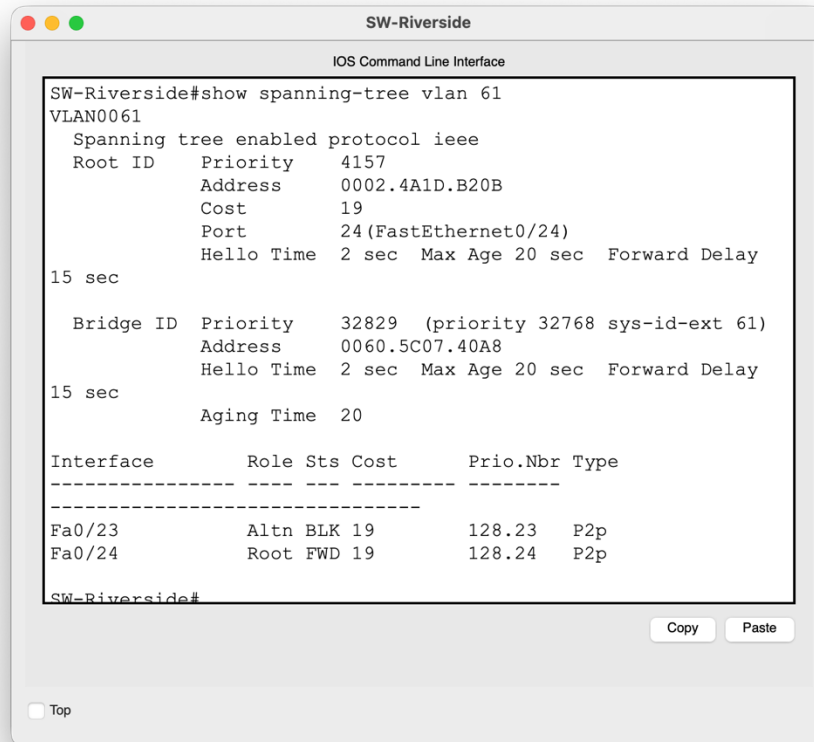
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14.9 Internet / ISP reachability



14.10 Convergence (before / after link failure)



SW-Riverside

IOS Command Line Interface

```
SW-Riverside#show spanning-tree vlan 61
VLAN0061
  Spanning tree enabled protocol ieee
    Root ID    Priority    4157
              Address     0002.4A1D.B20B
              Cost        19
              Port        24(FastEthernet0/24)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay
15 sec

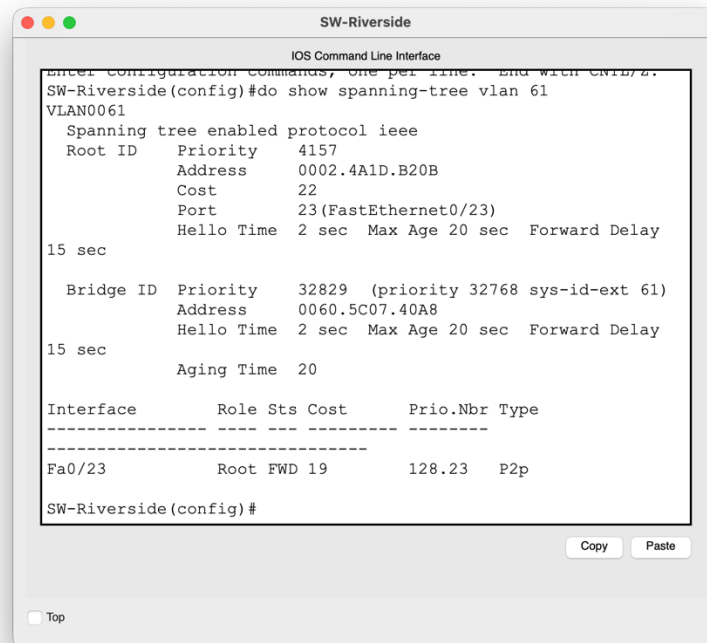
    Bridge ID  Priority    32829  (priority 32768 sys-id-ext 61)
              Address     0060.5C07.40A8
              Hello Time  2 sec  Max Age 20 sec  Forward Delay
15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/23         Altn BLK 19        128.23   P2p
Fa0/24         Root FWD 19        128.24   P2p

SW-Riverside#
```

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SW-Riverside

IOS Command Line Interface

```
Enter Configuration Commands, one per line. End with CNTRL-Z.
SW-Riverside(config)#do show spanning-tree vlan 61
VLAN0061
  Spanning tree enabled protocol ieee
    Root ID    Priority    4157
              Address     0002.4A1D.B20B
              Cost        22
              Port        23(FastEthernet0/23)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay
15 sec

    Bridge ID  Priority    32829  (priority 32768 sys-id-ext 61)
              Address     0060.5C07.40A8
              Hello Time  2 sec  Max Age 20 sec  Forward Delay
15 sec
              Aging Time  20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/23         Root FWD 19        128.23   P2p

SW-Riverside(config)#
```

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